

Rich Multimodal Inputs

Language

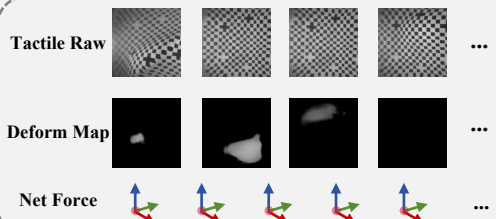
Screw the light bulb into the socket.

Vision



Head Camera Left Wrist Camera Right Wrist Camera

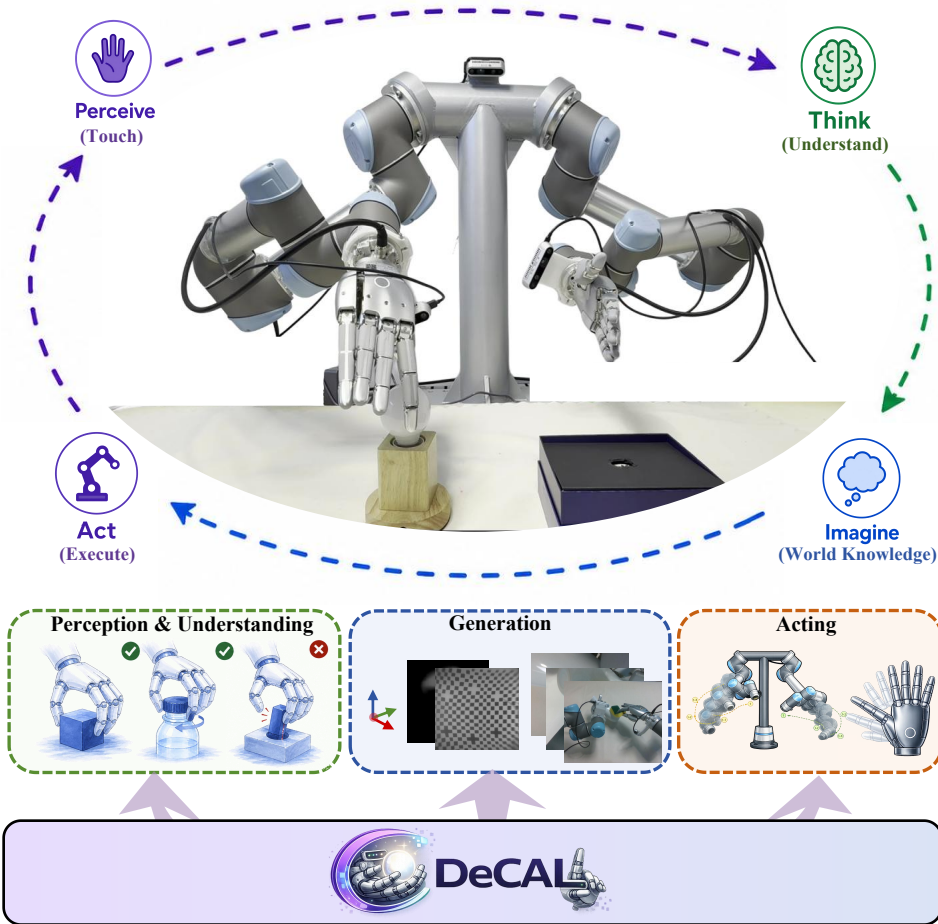
Tactile



- ✓ No Tactile Pretraining
- ✓ Adaptive Visuo-Tactile Fusion
- ✓ Visuo-Tactile Dynamics Modeling
- ✓ Unified Multimodal Experts

Touch, Think, Imagine, Act

A Unified Model for Dexterous Manipulation



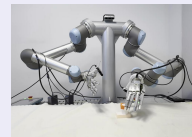
Contact-Rich Manipulation Tasks



Wipe Vase



Erase WhiteBoard



Assemble Parts



Twist Cap



Pipetting



Screw Light Bulb

Generalization



Unseen Background



Cluttered



Unseen Lighting



Unseen Object

Experimental Results Comparison

